



TEST PLAN

NutriVision

Abstract

This document describes the testing strategy, objectives, scope, test environment, and approach for validating the NutriVision web application. It provides the framework for planning, executing, and documenting manual testing activities throughout the project.

Prepared By
Alishba Rashid

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NutriVision – Smart Food Recognition & Nutrition Analyzer

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Document Type: Test Plan

Introduction

This Test Plan outlines the testing strategy, objectives, scope, resources, and testing approach for **NutriVision – Smart Food Recognition & Nutrition Analyzer**. The purpose of this document is to ensure that the application's features are tested thoroughly and meet the specified functional and non-functional requirements before release.

Test Objectives

The objectives of testing are to:

- Verify that all application features function according to the specified requirements.
- Validate user inputs and system responses.
- Ensure smooth navigation between application modules.
- Identify and document functional defects.
- Verify the application's usability and responsiveness across supported devices.
- Ensure the application provides a consistent and reliable user experience.

Test Scope

In Scope

The following modules will be tested:

- Landing Page
- Authentication
- Dashboard
- AI Food Analyzer
- Analysis Results
- Manual Food Search
- History
- Profile & Settings
- Navigation

- Form Validations
- Responsive Design

Out of Scope

The following are not included in this testing cycle:

- Real AI food recognition model
- Backend database integration
- Third-party API integrations
- Performance and load testing
- Security penetration testing

Test Strategy

Testing will be performed using **manual black-box testing** techniques. Test scenarios and detailed test cases will be executed to verify the application's functionality from an end-user perspective.

Testing activities include:

- Requirement Review
- Test Scenario Design
- Test Case Design
- Test Execution
- Defect Reporting
- Regression Testing
- Test Summary Preparation

Test Environment

Item	Details
Application	NutriVision
Platform	Web Application
Browser	Google Chrome (Latest Version)
Operating System	Windows 11
Testing Type	Manual Testing

Item	Details
Internet Connection	Required
Test Data	Mock User Data & Mock Nutritional Data

Test Types

The following testing types will be performed:

- Functional Testing
- UI Testing
- Form Validation Testing
- Navigation Testing
- Exploratory Testing
- Regression Testing
- Responsive Testing
- Usability Testing

Entry Criteria

Testing will begin when:

- Application development is completed.
- All required modules are accessible.
- Test environment is available.
- Test data is prepared.
- Requirements are documented.

Exit Criteria

Testing will be considered complete when:

- All planned test cases have been executed.
- Critical and high-priority defects have been resolved or documented.
- Regression testing has been completed for resolved defects.
- Test Summary Report has been prepared.

Test Deliverables

The following documents will be produced during testing:

- Software Requirements Specification (SRS)
- Test Plan
- Test Scenarios
- Test Cases
- Bug Reports
- Requirements Traceability Matrix (RTM)
- Test Summary Report

Risks & Mitigation

Risk	Mitigation
Mock AI responses may differ from real AI behavior	Testing will focus on implemented functionality rather than AI accuracy.
Requirement changes during testing	Update test documentation accordingly before execution.
Browser compatibility differences	Verify application on supported browsers where applicable.

Roles & Responsibilities

Role	Responsibility
QA Engineer	Prepare test documents, execute test cases, report defects, and verify fixes.
Developer	Resolve reported defects and implement required functionality.